Frazer Mills

+4407484867219 • psyfm5@nottingham.ac.uk • linkedin/FrazerMills • github/frazermills • website

Computer Science student and cloud enthusiast. Passionate about pursuing a career in a software or cyber organisation where I can explore a variety of technologies and their application within the software development lifecycle

EDUCATION

University of Nottingham | September 2023 - Present

- Computer Science BSc (Hons)
- First Year (78%): Computer Fundamentals (95%), Systems and Architecture (91%), Programming and Algorithms (82%), Intro to Software Engineering (81%)
- Awards: Witty Entrepeneurial Scholarship, 4th Highest (out of 76 groups) First Year Group Project with 91% average

Professional Certifications |

- AWS Certified Cloud Practitioner July 2024 - July 2027
- FreeCodeCamp Responsive Web Design May 2020

The Thomas Hardye School, Dorchester, Dorset | September 2018 - August 2023

- A Levels (Maths, Computer Science, Further Maths, Physics) + EPQ on Compilers AABC + A
- GCSEs + HPQ on AI 999999999887 + A*
- Awards: Isaac Stem Smart, Local University Scholars Programme, CyberFirst Advanced and Futures

PROFESSIONAL EXPERIENCE

Gemba Advantage | **Summer Intern** | *London, United Kingdom* June 2024 - August 2024

- Used AWS Cloud Practitioner Accreditation I achieved to rapidly replicate production system
- Developed a suite of security testing scripts that performed both Static and Dynamic Application Security Testing then provided a report with suggested improvements
- Wrote Terraform code to securely provision AWS services such as S3 buckets to store Terraform State with associated DynamoDB table inline with best practices
- Gained practical experience writing lambdas for document processing and interfacing with IoT core and MQTT

National Cyber Security Centre, GCHQ | **CyberFirst Advanced Programme** | *Remote* July 2022 - August 2022

- Lead team to win the Capture the Flag tournament
- Achieved highest individual score in number of tickets completed (out of over 130 students)
- Learnt about (with practical experience): Cryptography, Penetration Testing, Virtualisation, Vulnerability Detection

Hackathon | **Team Leader** | *University of Nottingham* November 2023

- Lead team to create a retro themed programming language (named c--) with a terminal emulator and text editor
- Wrote the Tokeniser, Lexer, Abstract Syntax Tree generator, and Parser for the language in C++ from scratch

EXTRACURRICULAR ACTIVITIES

School of Computer Science | **Student Mentor** | *University of Nottingham* July 2024 - Present

- Mentored a group of 15 first year students, providing academic and personal support
- Part of Nottingham Advantage award, took and additional 10 credit module to support my mentoring

Computer Science Society | **General Secretary** | *University of Nottingham* May 2024 - Present

- Facilited a partnership to bring internship opportunities at a non-profit start up to the socities' members
- Created template for the CompSoc instagram posts which have since gained

Programming Club | **Leader** | *Thomas Hardye School* September 2021 - June 2023

- Taught 7 students how to program in Python as well as libraries such as PyGame, NumPy, and SymPy
- Explained how my programming projects worked, such as Thermal and Gravitational Physics Simulations, an interpreter, and various games

PROJECTS

Educational Haskell YouTube Videos | July 2024 - Present

- Create frequent educational videos from my passion of programming in Haskell
- Cultivated an online community by responding to viewers' questions and to foster a deeper undertanding of functional programming

A Level Physics Simulator and Formula Calculator | September 2022 - May 2023

- Achieved full marks by building a particle physics simulation program using simple SUVAT to create animations and Verlet integration methods to simulate rigid body physics
- The program computes and displays various simulations and visualisations for my peers to view and interact with, so improving their ability to learn
- Created entire UI and Menu System Framework from scratch with only Pygame; custom button types created included: toggle buttons, text input boxes, resizable buttons, sliding buttons
- Built with Python and Pygame

Conway's Game of Life Visualiation | January 2021 - April 2021

- Full visual recreation of Conway's Game of Life with Python and Pygame using Object Oriented Design principals
- The project's codebase was forked by an external developer to create an indepedent but heavily inspired project which demonstrates the impact and adaptability of my work

ADDITIONAL INFORMATION

Other Awards | Senior Physics Challenge Ruby (top 180), Senior Maths Challenge Silver, UK Linguistics Olympiad Bronze

Technical Skills | AWS, Terraform, Python, C/C++, ARM Assembly, JavaScript, SQL, Unix/Unix-like, vim, emacs

Interests | Calisthenics, Chess, Compilers, Formula 1, Walking, Makoto Shinkai films